MSG No. 136.498 $P4'_2/mn'm$ [Type III, tetragonal]

Table 1: Wyckoff site: `2a`, site symmetry: `m.mm`

No.	position	mapping
1	$[0, 0, 0]$	`[1,2,3,4,5,6,7,8]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[9,10,11,12,13,14,15,16]`

Table 2: Wyckoff site: `2b`, site symmetry: `m.mm`

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	`[1,2,3,4,5,6,7,8]`
2	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[9,10,11,12,13,14,15,16]`

Table 3: Wyckoff site: `4c`, site symmetry: `2/m..`

No.	position	mapping
1	$\left[0, \frac{1}{2}, 0\right]$	`[1,2,5,6]`
2	$\left[\frac{1}{2}, 0, 0\right]$	`[3,4,7,8]`
3	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[9,10,13,14]`
4	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[11,12,15,16]`

Table 4: Wyckoff site: `4d`, site symmetry: `-4'..`

No.	position	mapping
1	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[1,2,13,14]`
2	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[3,4,15,16]`
3	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[5,6,9,10]`
4	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[7,8,11,12]`

Table 5: Wyckoff site: `4e`, site symmetry: `2.mm`

No.	position	mapping
1	$[0, 0, z]$	`[1,2,7,8]`
2	$[0, 0, -z]$	`[3,4,5,6]`
3	$\left[\frac{1}{2}, \frac{1}{2}, z + \frac{1}{2}\right]$	`[9,10,15,16]`
4	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2} - z\right]$	`[11,12,13,14]`

Table 6: Wyckoff site: `4f`, site symmetry: `m.2m`

No.	position	mapping
1	$[x, x, 0]$	`[1,3,6,8]`
2	$[-x, -x, 0]$	`[2,4,5,7]`
3	$\left[\frac{1}{2} - x, x + \frac{1}{2}, \frac{1}{2}\right]$	`[9,12,14,15]`

continued ...

Table 6

No.	position	mapping
4	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[10,11,13,16]`

Table 7: Wyckoff site: `4g`, site symmetry: `m.m2`

No.	position	mapping
1	$[x,\ -x,\ 0]$	`[1,4,6,7]`
2	$[-x,\ x,\ 0]$	`[2,3,5,8]`
3	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[9,11,14,16]`
4	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[10,12,13,15]`

Table 8: Wyckoff site: `8h`, site symmetry: `2..`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	`[1,2]`
2	$\left[\frac{1}{2},\ 0,\ -z\right]$	`[3,4]`
3	$\left[0,\ \frac{1}{2},\ -z\right]$	`[5,6]`
4	$\left[\frac{1}{2},\ 0,\ z\right]$	`[7,8]`
5	$\left[0,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[9,10]`
6	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}-z\right]$	`[11,12]`
7	$\left[0,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[13,14]`
8	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	`[15,16]`

Table 9: Wyckoff site: `8i`, site symmetry: `m..`

No.	position	mapping
1	$[x,\ y,\ 0]$	`[1,6]`
2	$[-x,\ -y,\ 0]$	`[2,5]`
3	$[y,\ x,\ 0]$	`[3,8]`
4	$[-y,\ -x,\ 0]$	`[4,7]`
5	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[9,14]`
6	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[10,13]`
7	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	`[11,16]`
8	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ \frac{1}{2}\right]$	`[12,15]`

Table 10: Wyckoff site: `8j`, site symmetry: `..m`

No.	position	mapping
1	$[x,\ x,\ z]$	`[1,8]`
2	$[-x,\ -x,\ z]$	`[2,7]`
3	$[x,\ x,\ -z]$	`[3,6]`
4	$[-x,\ -x,\ -z]$	`[4,5]`
5	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[9,15]`

continued ...

Table 10

No.	position	mapping
6	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	[10,16]
7	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	[11,13]
8	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	[12,14]

Table 11: Wyckoff site: 16k, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$[-x,\ -y,\ z]$	[2]
3	$[y,\ x,\ -z]$	[3]
4	$[-y,\ -x,\ -z]$	[4]
5	$[-x,\ -y,\ -z]$	[5]
6	$[x,\ y,\ -z]$	[6]
7	$[-y,\ -x,\ z]$	[7]
8	$[y,\ x,\ z]$	[8]
9	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	[9]
10	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	[10]
11	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	[11]
12	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	[12]
13	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	[13]
14	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	[14]
15	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	[15]
16	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	[16]